

Headline XIII: the general-dimensional phase-dressed leading asymptotic

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Abstract

Unit 203 wraps the deterministic core of programme A (unit 202) in the paper's notation. With $p = 2\lambda$, J the minimiser set, $m = |J|$, π the projection to the face $u_J = 0$, and $J_p(a) = \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta a s} ds$ (**phaseMoment** p a):

$$\frac{\int_{[0,1]^d} \eta(u) u^h e^{-\beta N^2 u^{2k} + \beta N u^k \xi(u)} du}{N^{-p} (\log N)^{m-1}} \longrightarrow \frac{1}{(m-1)! \prod_{i \in J} k_i} \times \int_{[0,1]^d} \eta(\pi u) J_p(\xi(\pi u)) \prod_{i \notin J} u_i^{h_i - p k_i} du$$

(**headline_phase_leading**); when all ratios are equal the limit is the corner value $\eta(0) J_p(\xi(0)) / ((d-1)! \prod_i k_i)$ (**headline_phase_leading_equal**), which for $d = 2$ is exactly the paper's Taylor-tree leading coefficient $A_p = y_{00} J_p(x_{00}) / (k_1 k_2)$. The zero-phase regression **phaseCoeff_zero_phase** identifies **phaseCoeff**($\xi = 0$) with 2^{m-1} times the amplitude coefficient of unit 186. Zero sorry/axiom; 9080 jobs.

1 The things formalised

```
theorem multCount_eq_card : multCount 1 = (univ.filter fun i => i = 1).card
theorem prod_ite_half_div : i, (if ratioExp h k i = 1 then 1/(2 k i) else 1)
  = (1/2) ^ multCount (ratioExp h k) 1 * i, (if ratioExp h k i = 1 then 1/(k i)
    else 1)
theorem two_pow_mul_faceNorm (hm : 1 multCount ...) :
  2 ^ (m - 1) * 2 * faceNorm h k 1 = 1 / ((m - 1)! * i, if ratioExp h k i = 1
    then k i else 1)
theorem paperKernel_eq_quadKernel :
  exp (-( N^2 u^(2k)) + (N u^k) u) = quadKernel (u) (N u^k)
theorem headline_phase_leading (hk) (hl) (h) (hmin) (hatt) (h) (h) :
  Tendsto (fun N => ( u in unitBox (m+1), u * (( u^h) * exp (-( N^2 u^(2k))
    + (N u^k) u)))
    / (N^(-(2l)) * log N ^ (multCount (ratioExp h k) 1 - 1))) atTop
  ( (1 / ((multCount ... - 1)! * i, if ratioExp h k i = 1 then k i else 1)
    * u in unitBox (m+1), ( (faceProj h k 1 u) * phaseMoment (2l) (
      (faceProj h k 1 u)))
    * residualWeight h k 1 u))
theorem headline_phase_leading_equal (hratio : i, (h i + 1)/(2 k i) = 1) ... :
  Tendsto (fun N => ... / (N^(-(2l)) * log N ^ m)) atTop
  ( ( 0 * phaseMoment (2l) ( 0) / (m! * i, k i)))
```

```

theorem phaseCoeff_zero_phase (hl) (h) :
  phaseCoeff h k l (fun _ => 0) = 2 ^ (multCount (ratioExp h k) l - 1) *
    amplitudeCoeff h k l

```

2 Notation correction

Units 201–202 (retros and docstrings) called $\int_0^\infty s^{p-1} e^{-\beta s^2 + \beta a s} ds$ “the paper’s $S_{p/2}(a)$ ”. The paper’s fluctuation function is $S_\lambda(a) = \int_0^\infty t^{\lambda-1} e^{-\beta t + \beta a \sqrt{t}} dt$, and $t = s^2$ gives $S_{p/2}(a) = 2 \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta a s} ds$. So `phaseMoment p a` is $J_p(a) = S_{p/2}(a)/2$, the object that actually appears in the paper’s $A_p = y_{00} J_p(x_{00})/(k_1 k_2)$. The docstrings are corrected in this branch; the Lean statements were never affected (they name `phaseMoment`, not `S`). The constant check against the paper: for $d = 2$ with equal ratios the wrapper gives $\eta(0) J_p(\xi(0))/(1! k_1 k_2)$, which is A_p with $y_{00} = \eta(0)$, $x_{00} = \xi(0)$, and reduces at $\xi = 0$ to $\eta(0) \Gamma(p/2) \beta^{-p/2} / (2 k_1 k_2) = 2 \cdot \Gamma(\lambda) \beta^{-\lambda} / (1! \cdot 2 k_1 \cdot 2 k_2)$, the “twice the monomial mixed constant” of unit 193.

3 Mathematics

The paper constant: $2^{m-1} \cdot 2 \cdot \frac{1}{(m-1)!} \prod_{i \in J} \frac{1}{2k_i} = \frac{2^m}{(m-1)!} \cdot \frac{1}{\prod_{i \in J} k_i}$, via $\prod_i (\text{if } i \in J \text{ then } \frac{1}{2k_i} \text{ else } 1) = (\frac{1}{2})^{|J|} \prod_i (\text{if } i \in J \text{ then } \frac{1}{k_i} \text{ else } 1)$ and $|J| = \text{multCount } (\text{Finset.sum_boole}, \text{Finset.prod_ite}, \text{Finset.prod_const})$. In the equal-ratio case $\pi \equiv 0$ and the residual weight is 1, so the face integral is the constant $\eta(0) J_p(\xi(0))$ times the unit volume.

4 Lean notes

- `rw [if_pos (hratio i)]` fails when the `if` condition is the folded `ratioExp h k i = 1`: give the proof the folded type, `if_pos (show ratioExp h k i = 1 from hratio i) (defeq, but rw matches syntactically)`.
- `Finset.prod_ite` rewrites the first matching product; when two products of `ifs` are present, isolate the intended one in a `have` first.
- `ring` proves $2^m \cdot (1/f \cdot ((1/2)^m \cdot P^{-1})) = (2^m (1/2)^m) \cdot (1/f \cdot P^{-1})$ without knowing $2^m (1/2)^m = 1$; do the cancellation as a separate rewrite so no nonvanishing hypothesis on $\prod k_i$ is needed.

5 Status

Programme A (deterministic core) is complete in five units (199–203). Next: mirror entry `eq:normal_moment_phase_lean`, fidelity review v9 (units 198–203), report v2.